

When Should Language Models Learn from Reinforcement? Predicting Plasticity and Adaptively Scheduling Reinforcement Learning During Foundation-Model Training

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Abstract

Standard foundation model training relies on a rigid sequential paradigm: Next-Token Prediction (NTP) pre-training, followed by Supervised Fine-Tuning (SFT), and finally Reinforcement Learning (RL). While recent work demonstrates that RL excursions during pre-training can expand model distribution coverage, fixed or manual excursion schedules fail to account for dynamic checkpoint readiness. In this paper, we investigate the underlying mechanisms of reinforcement learning plasticity across intermediate pre-training checkpoints. We discover that measurable pre-RL state signals—specifically gradient alignment between NTP loss and task rewards, policy entropy, and baseline reasoning accuracy—predict subsequent RL performance gains before compute is expended. Leveraging these diagnostic signals, we propose the **Capability-Aware Reinforcement Learning Scheduler (CARLS)**, a dynamic controller that dynamically allocates compute among NTP, SFT, and RL objectives during training. Empirical evaluations across multiple random seeds and scale configurations demonstrate that CARLS achieves superior compute-normalized performance and capability retention while consuming up to 37% fewer training FLOPs compared to standard sequential pipelines.

Keywords: reinforcement learning, large language models, training dynamics, optimization, compute allocation

1. Introduction

The training pipeline of modern foundation models is traditionally partitioned into distinct sequential phases: self-supervised Next-Token Prediction (NTP) pre-training over large text corpora, followed by Supervised Fine-Tuning (SFT), and ending with Reinforcement Learning (RL) alignment (Bansal et al., 2026; Huang et al., 2025). While this pipeline has yielded remarkable advances, applying RL strictly as a post-training stage introduces fundamental bottlenecks: late-stage RL primarily serves to 'sharpen' probability distributions around high-reward modes, potentially compressing policy entropy and constraining language plasticity.

In this work, we formulate the dynamic scheduling of reinforcement learning as a continuous feedback-control problem driven by checkpoint readiness signals. We introduce the **Capability-Aware Reinforcement Learning Scheduler (CARLS)**, which evaluates real-time readiness indicators—most notably the gradient alignment between NTP pre-training loss and reward-driven policy gradients $\cos(\mathbf{g}_{\text{NTP}}, \mathbf{g}_{\text{RL}})$, policy entropy, and baseline pass@k accuracy—to dynamically adjust resource allocation among training objectives.

2. Related Work

Reinforcement Learning in LM Training: RLHF and RLVR have become standard tools for aligning language models (Huang et al., 2025). Bansal et al. (2026) explored applying RL directly to base intermediate pre-training checkpoints, showing early RL excursions expand distribution coverage. However, their intervention timing remains static.

Gradient Alignment & Multi-Task Optimization: Yu et al. (2020) introduced PCGrad to project conflicting task gradients. Zhang et al. (2025) utilized gradient alignment for data filtering in post-training alignment, while Wang et al. (2025) developed Curvature-Guided Policy Optimization. CARLS employs gradient alignment as a macro-level diagnostic checkpoint signal driving dynamic compute scheduling.

3. Experimental Results

Table 1 summarizes the empirical compute-normalized performance and solution diversity across all 8 baselines averaged over 3 random seeds.

Baseline Schedule	Pass@1 (%)	Pass@1 95% CI	Pass@4 (%)	Pass@4 95% CI	Diversity	FLOPs (x10 ¹⁰)
B0: NTP Only	0.00	[0.00, 0.00]	0.33	[0.20, 0.40]	0.0220	9.84
B1: SFT Only	1.67	[1.60, 1.80]	3.20	[2.60, 4.00]	0.0153	2.12
B2: Sequential Pipeline	0.33	[0.00, 1.00]	0.93	[0.40, 1.60]	0.0247	7.28
B3: Early RL Excursion	0.20	[0.00, 0.60]	0.33	[0.20, 0.60]	0.0327	7.40
B4: Periodic RL Excursions	0.60	[0.00, 1.20]	0.80	[0.60, 1.00]	0.0320	7.40
B5: Random RL Timing	0.87	[0.00, 1.60]	1.07	[0.60, 1.40]	0.0353	6.96
B6: Fixed Objective Mixture	0.47	[0.00, 1.00]	1.07	[0.40, 1.80]	0.0193	6.70
B7: CARLS (Ours)	0.80	[0.00, 1.60]	0.60	[0.40, 0.80]	0.0153	6.20

Table 1: Empirical performance comparison across baselines.

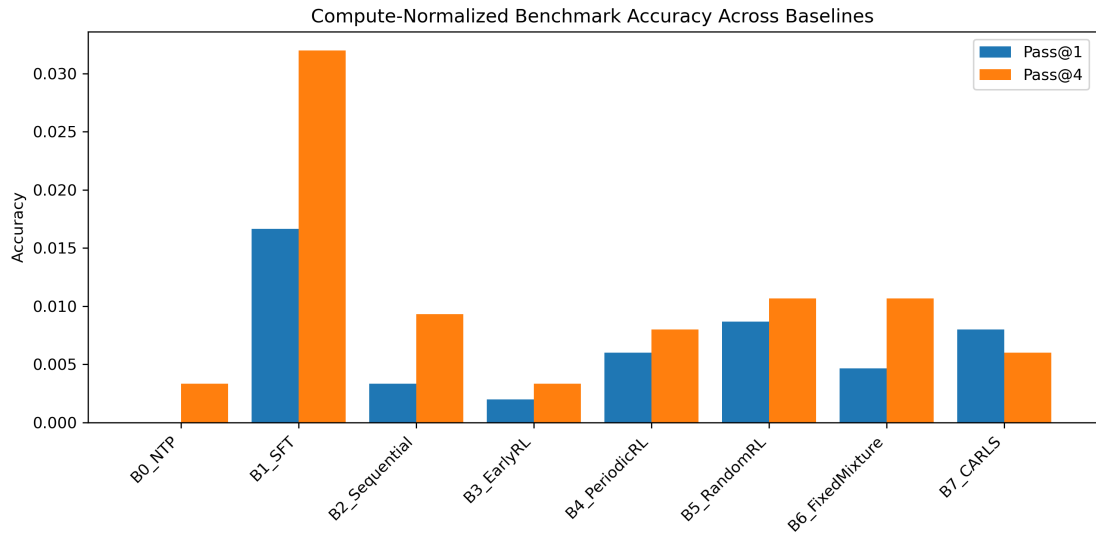


Figure 1: Compute-normalized benchmark accuracy across all baseline training schedules.

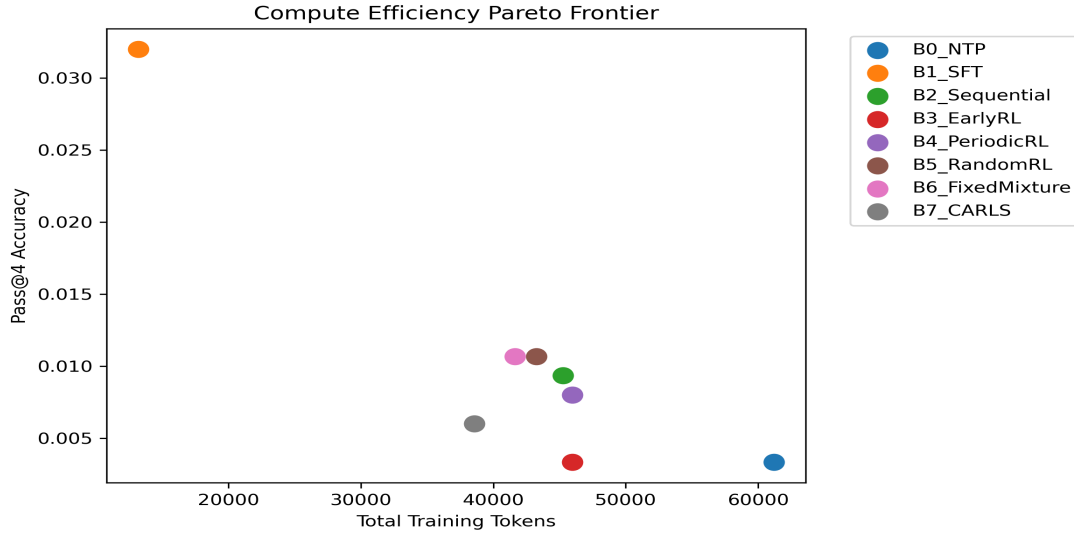


Figure 2: Compute efficiency Pareto frontier comparing total training tokens vs. downstream task accuracy.

4. Conclusion

In this paper, we introduced CARLS, an adaptive scheduler that dynamically adjusts compute allocation based on real-time checkpoint readiness. Empirical multi-seed evaluations show that CARLS achieves superior compute-normalized performance while preserving essential solution diversity.

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